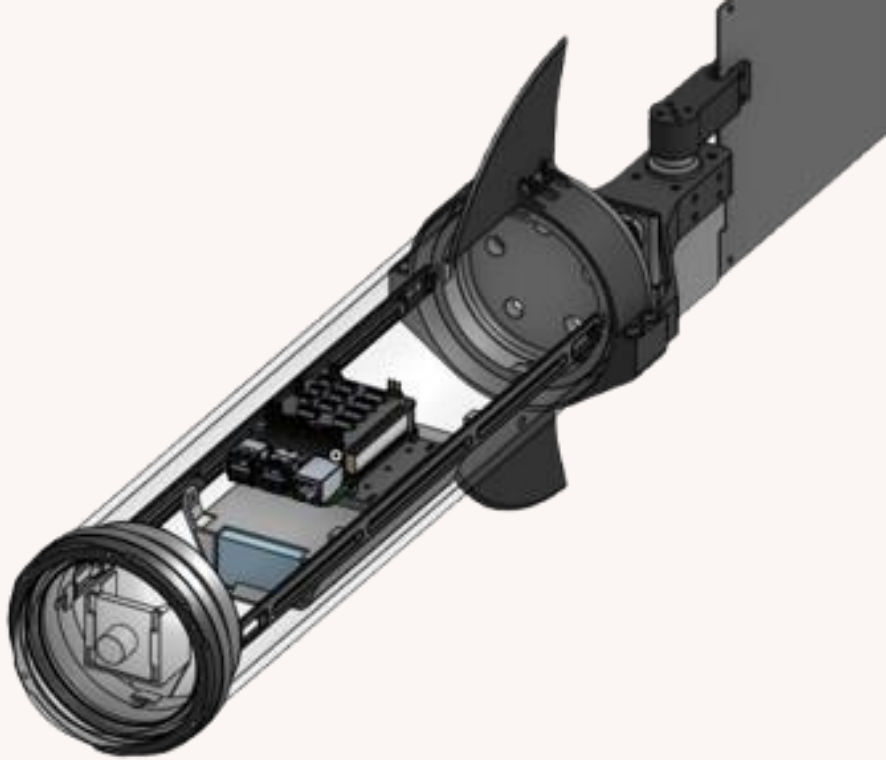
Project Presentation

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4 Teams – 1 Project – 40h



Complete conception of a micro-AUV:

- Mechnical conception,
- Electrical Conception,
- Software development.

3D realistic Unity-ROS2 simulation development:

- 3D environment and physics simulation,
- Sensor and actuator,
- ROS2 integration and communication.

Brazilian Team

Irys Team

Baby shark Team

Team United

AUV Conception

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Electronical Conception







What is provided ?

- Raspberry pi 4
- Navigator hat
- Power module & BEC
- Battery
- Fin servo
- Mass servo
- Depth sensor
- Camera
- (GNSS)
- (Pinger)



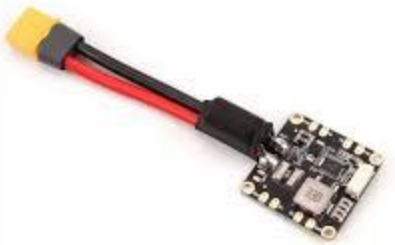




Objective :

- Understand the role of each components
- Draw the electronical diagram
- Put the electronical assembly together
- Realize the corresponding ROS2 architecture







What is provided ?

- Onshape CAD :
  - Main assembly
  - Caps and Flanges
  - Mounts for :
    - Raspberry
    - Power module & BEC
    - Battery
    - Camera
- Components :
  - Tube
  - Dome
  - Rails

Objectives :

- Design the remaining parts:
  - Fin mechanism
  - Sliding mass mechanism
- Build, print and put the assembly together

Onshape CAD :

- Create Onshape student account with @ensta.fr
- Search for "ENSTA_ROB Ingénierie Système 2026" on public files









Workshops :

- SLA 3D printing
- Tapping
- Tube cutting
- Mass balancing
- Underwater servo assembly

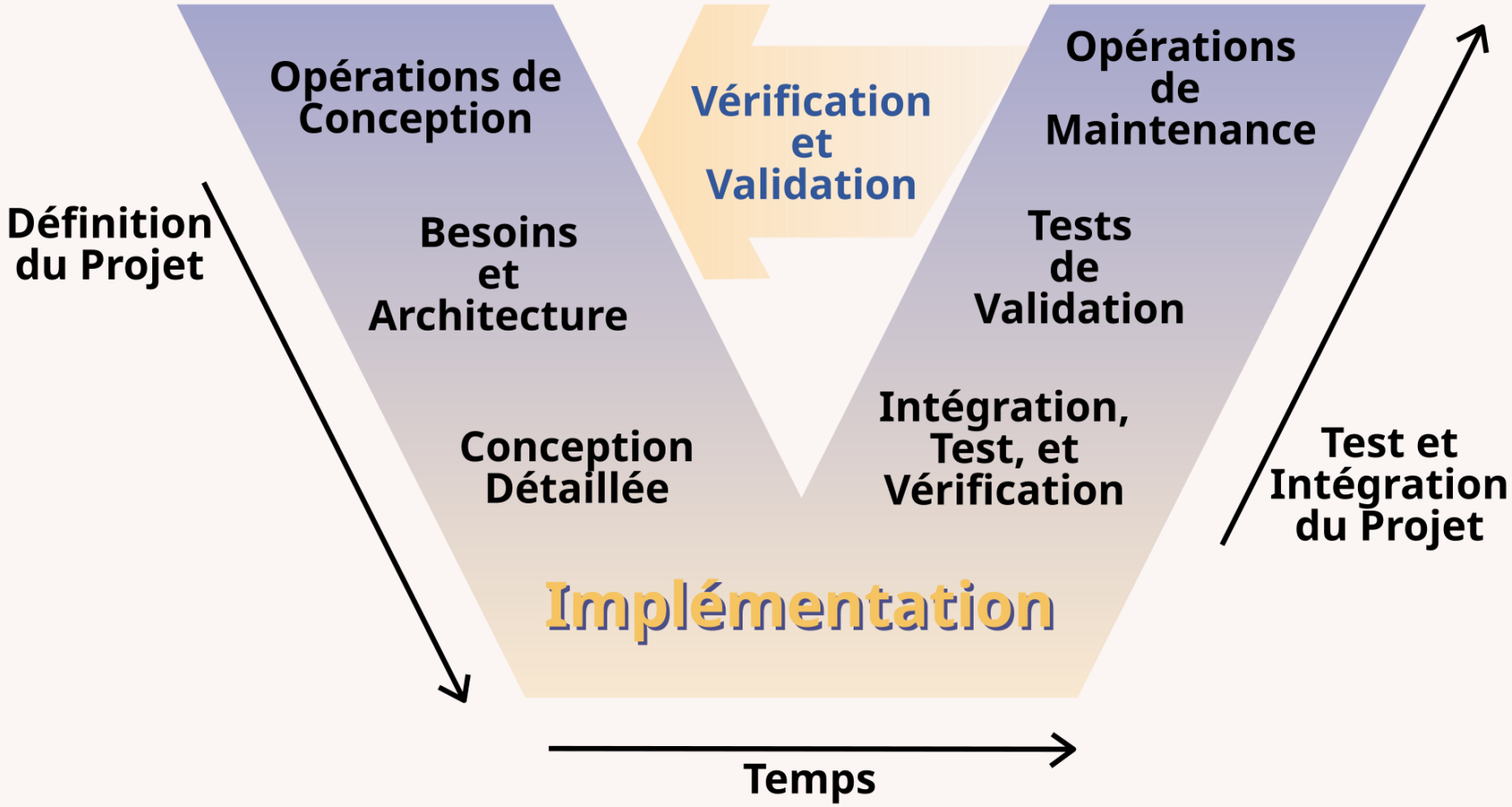
System Engineering

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The V-Model: A Tool for Large Organizations



Useful in large, hierarchical organizations (Naval Group, Thales, Airbus).

➡ **Not suitable for this course:** 6 students with no hierarchical relationships.

Roles in the Agile Scrum Framework

Product Owner (PO)

- Ensures project success.
- Manages client meetings.
- Prioritizes the Product Backlog.
- Accepts or rejects deliverables.
- Leads Sprint Planning.




Scrum Master

- Facilitates the team, not a project manager.
- Leads the 15-minute Daily Scrum.
- Ensures team efficiency and cohesion.
- Protects the team from external disruptions.
- Facilitates internal communication.



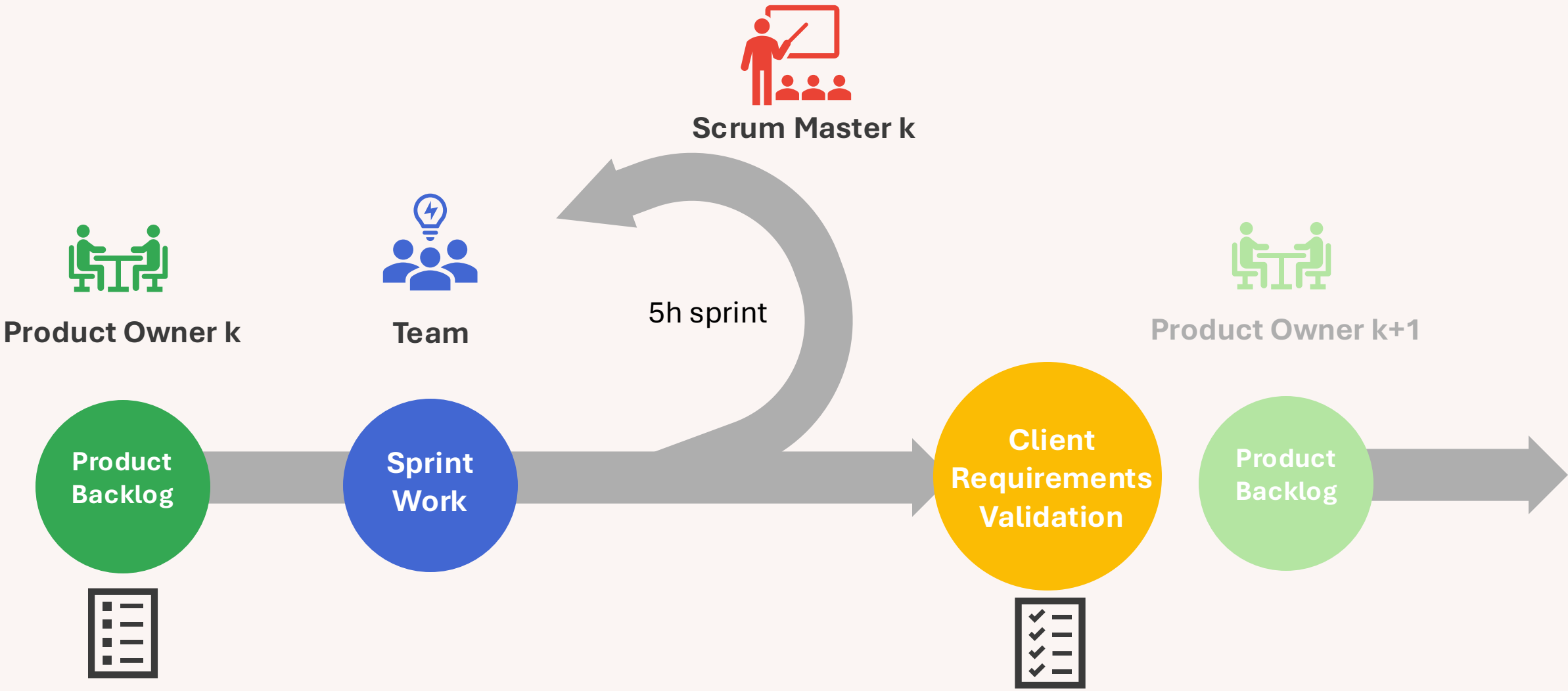

Development Team:

- Self-organizing, with no internal hierarchy.
- Implements technical solutions.
- Maintains detailed project specifications.
- Delivers a working product increment.





Agile Methodology illustration



Github & README: Sharing and Showcasing Your Work

Make Videos!

- Simulator demonstration video.
- Videos documenting the design, development, and testing of your AUV.

Organise your work:

- 10-15 min
- 1 Product Owner
- 1 Scrum Master

Document everything:

- Code documentation: Explain how to install, configure, and run your code.
- System architecture: Provide electrical and mechanical diagrams.
- Software architecture: Describe the software structure and how its components interact.

Client Meeting Organization

	RDV1	RDV2	RDV3	RDV4	RDV5	RDV6
Pedro						
Pedro Henrique						
Luiz Felipe						
Matheus						
Daniel						
Leticia						

	RDV1	RDV2	RDV3	RDV4	RDV5	RDV6
Florian						
Anatole						
Vivien						
Thomas						
Guilhem						
Yoann						

Scrum Master

Product Owner

	RDV1	RDV2	RDV3	RDV4	RDV5	RDV6
Noé						
Clément						
Gaspard						
Eliaz						
Lancelot						
Prénom Nom 6						

	RDV1	RDV2	RDV3	RDV4	RDV5	RDV6
Kadijatou						
Elodie						
Tom						
Alexandre						
Luiza						
Matys						

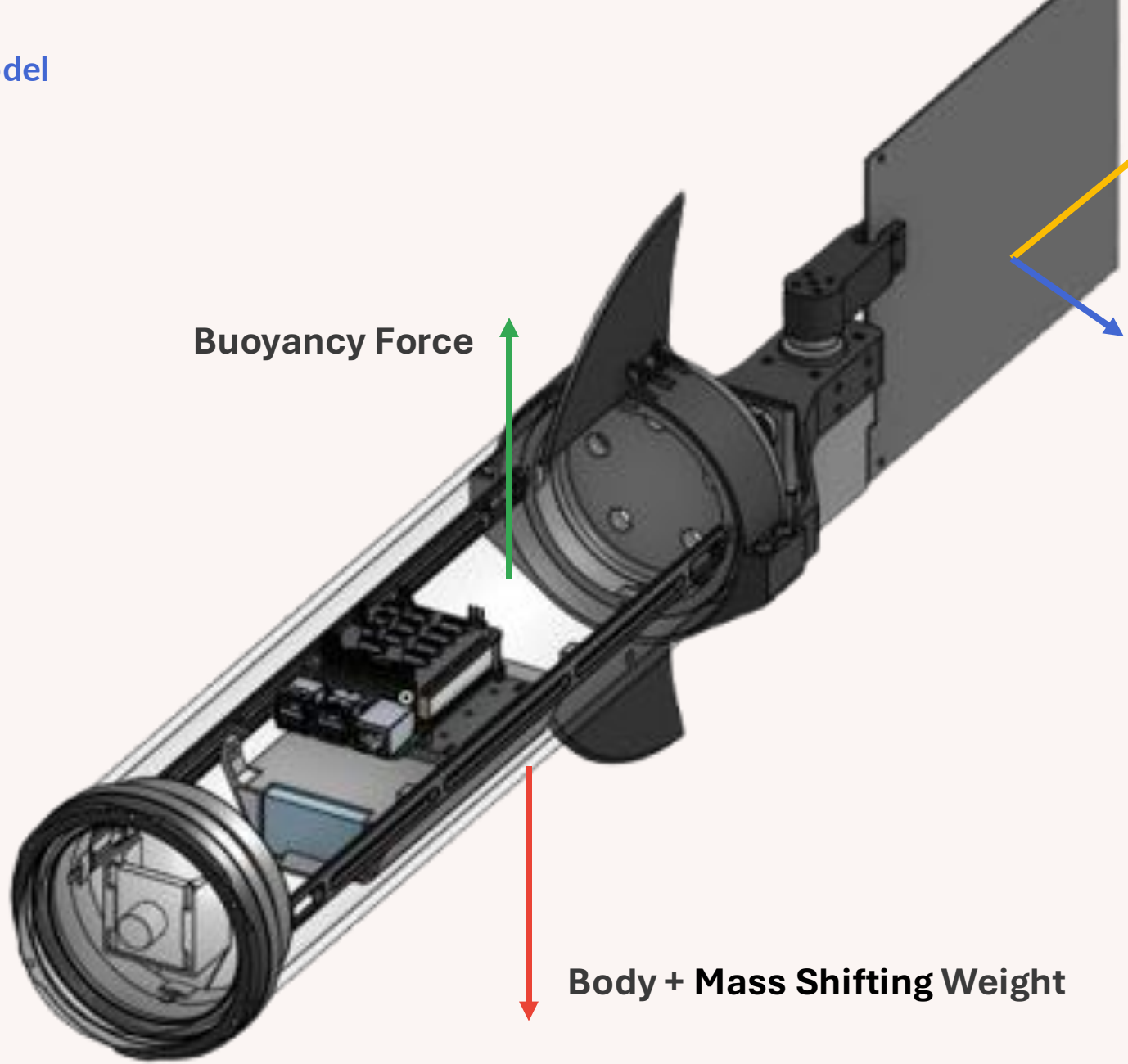
Google sheet [link]

3D Simulation

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Physical Model



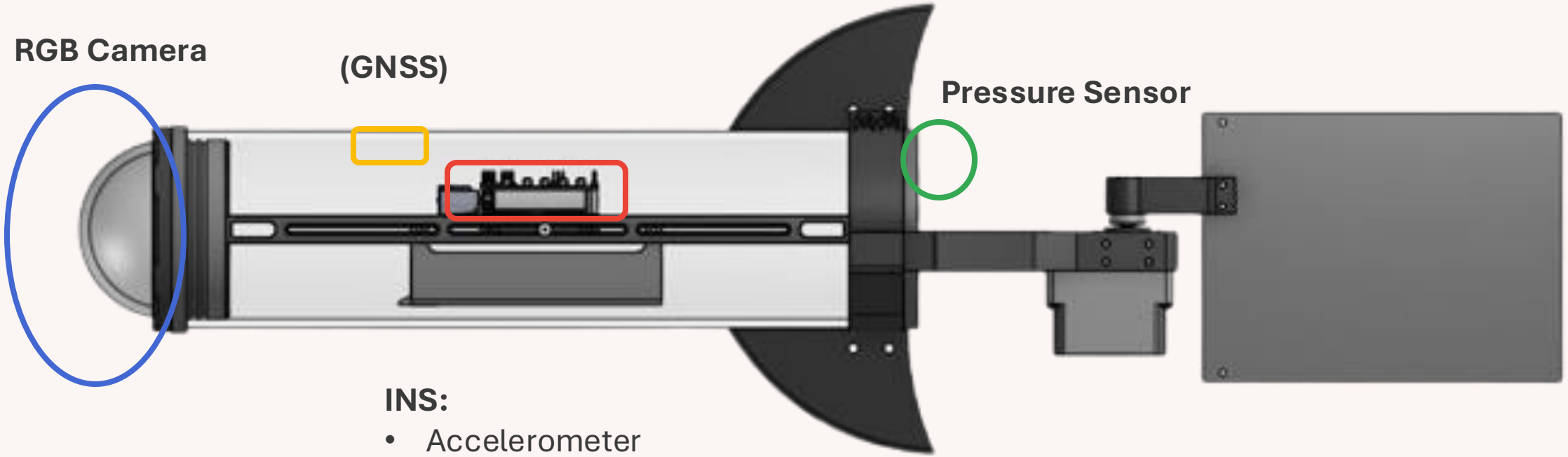
Fin Thrust

Fin Lift

Buoyancy Force

Body + **Mass Shifting** Weight

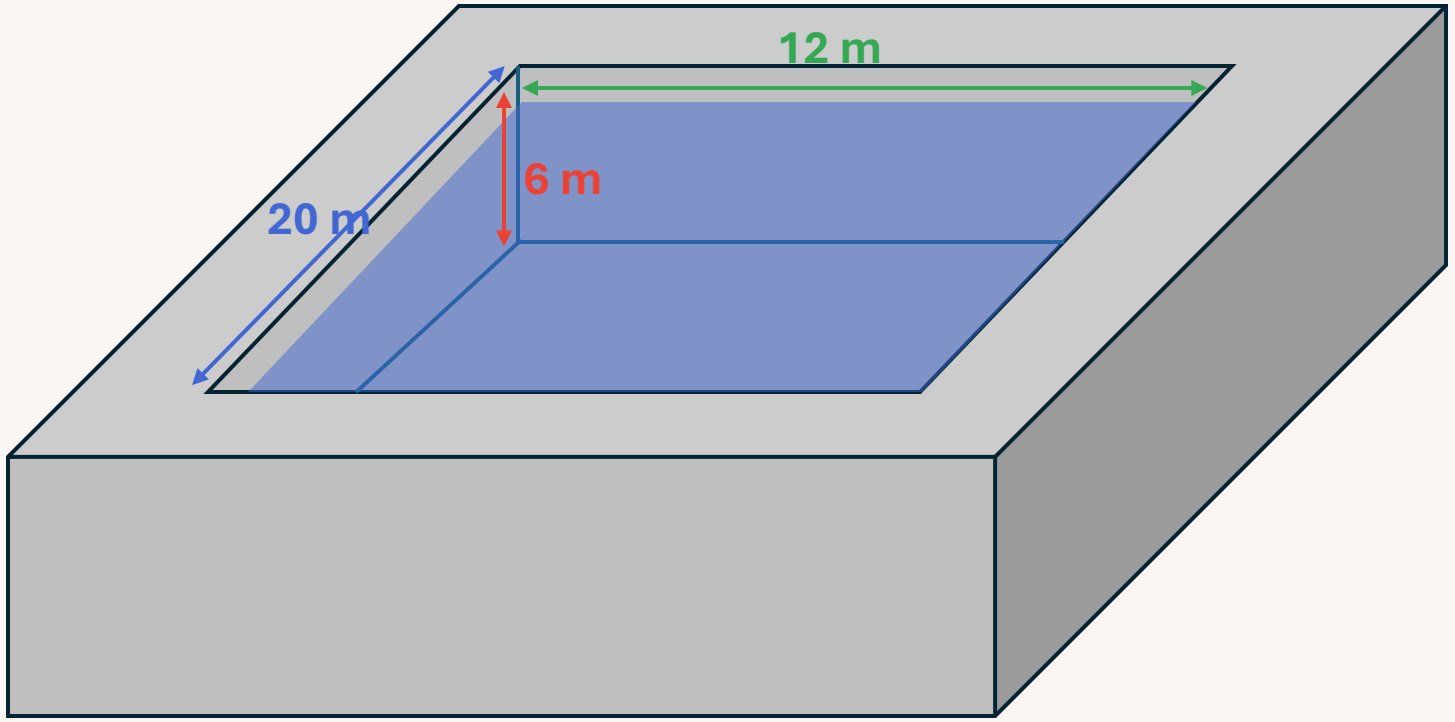
+ Linear and angular
Hydrodynamic drags



INS:

- Accelerometer
- Gyroscope
- Magnetometer

3D environment



Summary of Expected Simulator Deliverables

Physics
- Weight,
- Buoyancy force,
- Fin lift,
- Fin thrust,
- + linear and angular drags.

Sensors
- RGB camera,
- INS : accelerometer, magnetometer, gyroscopes,
- Pressure sensor,
- (GNSS)

3D model
- On Ubuntu 24.04, you can use free, open source online software Onshape

3D scene
- Terrain
- Environment
- (Skybox, props ...)

And high-level intelligence to have heading following, depth regulation etc..
- Navigation
- Guidance
- Control

Summary of Expected Conception Deliverables

Mechanical
- Waterproofing,
- Buoyancy and weight balancing,
- 3D pieces design and printing,
- Real conditions tests.

Electronical
- Electrical schematic

Software
- ROS2 Drivers
- ROS2 architecture

Videos
- To show your work

Summary of Expected Clients Deliverables

Study of project cost
- Hardware & Components
- Human hours working
- Proposition of new components

Must be agreed upon and verified by every team member.

A presentation at every client meeting
- Powerpoint template exemple can be found here

README : a great showcase of your work
- Presentation of everything you have done,
- Video of the simulator,
- Video of AUV basin’s tests.

Thank you for your attention

Do you have any question?

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